2 Digital literacy: Staying safe online

You will learn

- how to recognise the risks and dangers on the internet
- how to avoid risks and dangers on the internet
- how to use internet content responsibly.



In Student Book 5 you learned what it means to be a responsible internet user. Not everyone who uses the internet behaves responsibly. There are some people who make the internet a dangerous place. There are criminals who attempt to steal from you. Others try to install malicious software on your computer. There are bullies who use the internet to frighten and threaten people.

In this unit you will learn about the risks that the internet brings into our lives. You will learn the steps you need to take to guard against these risks.

Unplugged

Answer these two questions individually. How much do you agree with each statement? Choose the answer that is true for you.

"I feel safe when I am online."	"I am confident that I know how to stay safe online."
1 I feel very safe.	**1** I am very confident.
2 I feel quite safe.	2 I am quite confident.
3 I feel neither safe nor unsafe.	3 I am neither confident nor unconfident.
4 I sometimes worry I am at risk.	4 I don't know very much about staying safe.
5 I think the internet is a dangerous place.	5 I know nothing about staying safe online.

Collect the responses of everyone in the class. Create a graph to present your class's responses.

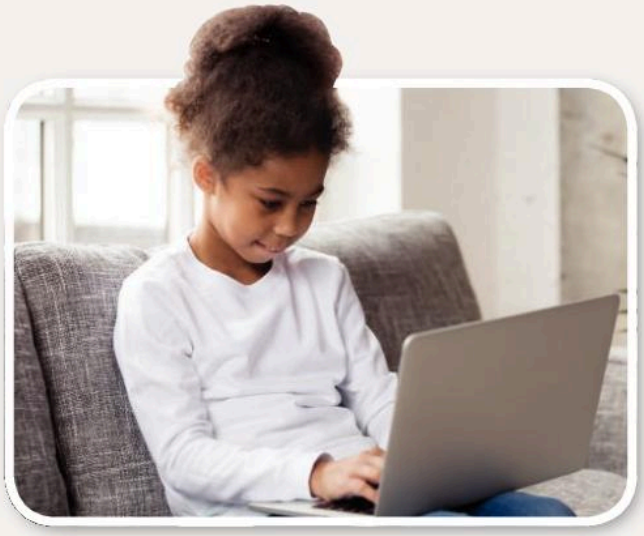
Learning outcomes: Use content from online sources responsibly; Explain risks associated with internet use; Discuss how data may be collected when working online

Cybercrime

Identity theft

Hacking

Malware



Virus

Cyberbully

Ransomware

Talk about...

Some of the threats you face when using the internet are shown above. You will learn more about these threats in this unit, but you may already have heard about some of them. Share what you already know about these threats with your group.

Which of the threats do you worry about most? What do you do to stay safe online?

Did you know?

By 2021 the cost of internet crime is expected to be $6 trillion a year. Businesses across the world will spend $1 trillion to protect themselves against criminals who use the internet.



cybercrime
hacking malware cyberbully
anti-virus software internet shopping
e-commerce secure site cookie
intellectual property copyright
plagiarism firewall

2.1 Collecting data online

In this lesson

You will learn:

- how data is collected online
- how cookies are used on websites.

Finding information online

There are many reasons to look for information. You might do a web search to find information for a geography project. You might look for reviews online to help you decide which model of smartphone is best for you.

In the modern world, the information you need is usually on the internet. When you use the internet you also give away information. In this lesson you will learn about the ways in which websites collect data about you.

Online registration

Websites such as social media and internet shopping sites usually ask you to register before you can use them. **Registration** means becoming a member of a website. You fill in an online form and provide information about yourself to the owner of the site.

In return for registering, you receive some benefits. For example, you might be able to:

- read pages that are hidden from non-registered users
- leave messages in chat rooms and on message boards
- add content
- get email updates when new content is added to the site
- download software.



Which information should you give?

The owner of a website should only collect the information they need, such as a username and password. They may also need contact details such as an email address. Some websites ask for optional information such as your telephone number or birthday. You don't have to give optional information to register. Always think about what information you need to provide when you complete forms online.

Activity

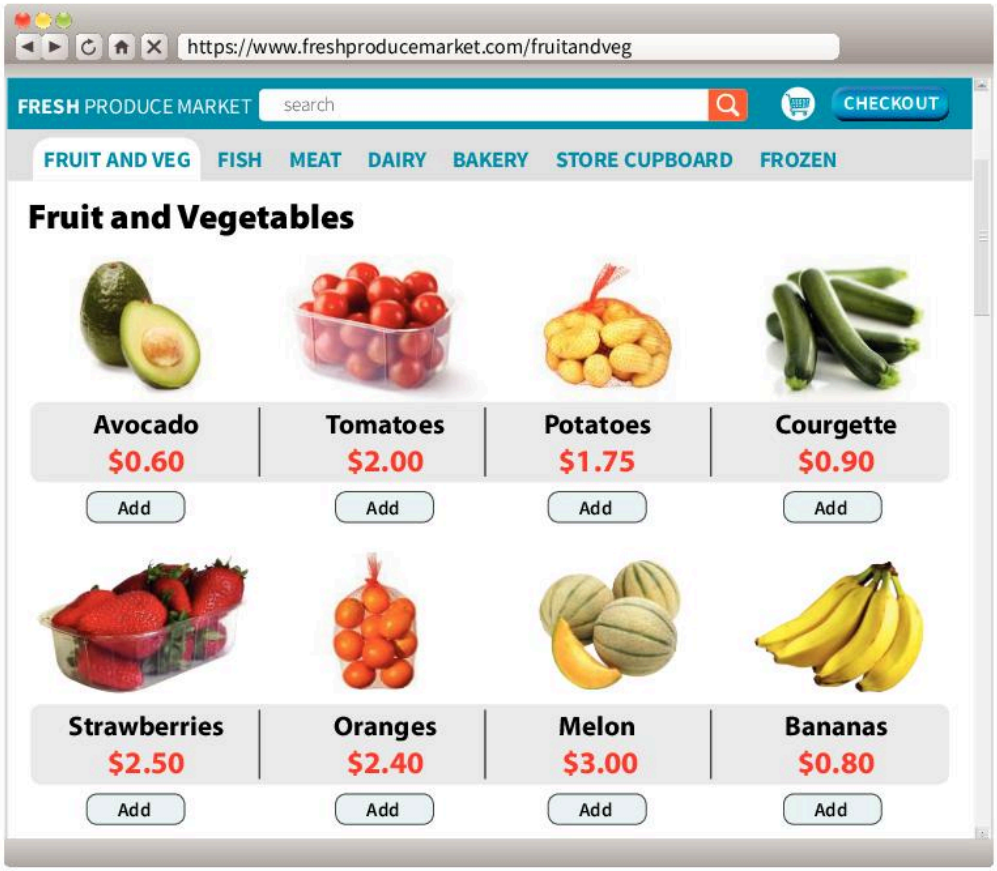
Imagine you have designed a new website about your favourite computer game. Your website has a chat room so that friends can discuss the game and share tips on how to play. What information will you collect from your friends when they register? Design a registration form to use on your website.

Online shopping

Buying online is called internet shopping or **e-commerce**. Internet shopping sites collect data when people buy things. Some of the data that shopping sites collect is personal data, including:

- bank details – so we can pay for the items
- address and other contact details – so the company can deliver the items.

Website owners must keep personal details safe. A criminal can use a person's bank details to steal money from their bank account. A criminal can use personal details such as a person's address to impersonate (pretend to be) the person and then commit crimes in their name.



Internet shops use **secure sites** to protect data. A secure site **encrypts** information that is sent over the internet. Encrypted data is coded. If a criminal steals the data they will not be able to read or use the data.

Is this site secure?

When you are browsing the web, there are two clues that tell you a website is secure.

There is a small locked padlock symbol.

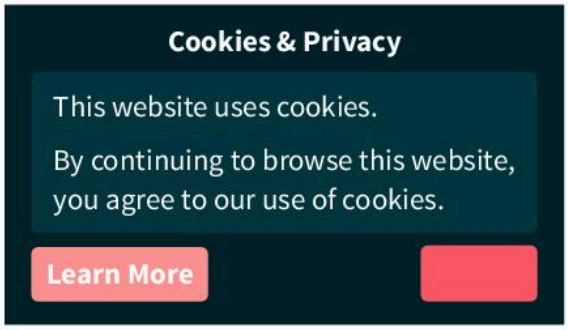


The URL of a secure site starts with https://. The 's' stands for secure. It tells you that data is encrypted when it is sent over the internet.

Never send information across the internet if the website is not secure. However, not all secure sites are safe to use. The safest way to send information is to use a secure site that you have used before and can trust.

Cookies

A **cookie** is a small file that is stored on your computer when you visit a website. A cookie saves information about the way you use web pages. The website uses this information to improve your experience of using the website.



Types of cookies

There are four types of cookie.

- **Essential cookies** are needed to make a website work as designed. On an internet shopping site, the shopping basket function needs cookies to make it work properly.
- **Performance cookies** collect information about how you use a website. The owner of a website uses this information to improve the performance of the website.
- **Functionality cookies** record what you do on a web page. This information is used to personalise the web page for you.
- **Advertising cookies** record what you look at online. The information is used to personalise the adverts displayed on a web page.

What are cookies used for?

Cookies are used for two main reasons:

- Cookies make it easier to use websites. For example, if you complete a form on a website, the information is saved in a cookie. This means that the website can automatically complete the form the next time you use it.
- Cookies are used to personalise a website. For example:
  - A cookie can store your location. That information can be used to make sure you see weather reports and events that are local to you.
  - Advertisers use cookies to make sure they show you adverts for events and products you are interested in. They can collect this information from your browsing history.
  - The news stories that you see can be customised to match your interests.



Did you know?

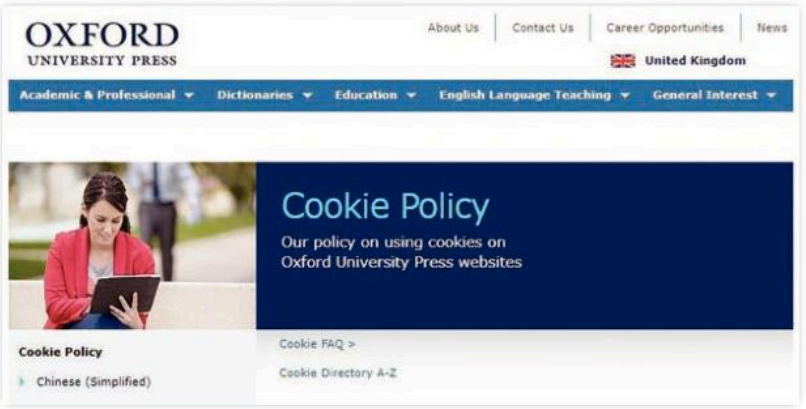
The first internet shop was called NetMarket. It made its first sale in 1994. In 2019, 1.92 billion people were expected to buy something on the internet.

Cookies and the law

The information that cookies collect is often very valuable to advertisers. Some website owners sell the information to advertisers and other organisations such as political parties. Some cookies have been developed to collect valuable information that can then be sold. These are called tracking cookies.

Some people worry that tracking cookies can be used to influence our opinions and affect the way we vote in elections.

The governments in some countries have passed laws about the use of cookies. Website owners must make it clear what cookies their sites use and what they use the cookies for. These laws make it easier for people to decide which cookies to accept and which to reject.



Activity

Draw a table with two columns like the one shown here.

Good uses of cookies	Bad uses of cookies

In the first column list all the ways that cookies make using the internet better. In the second column list all the bad things that cookies are used for.

Extra challenge

Search the web to find out more information about cookies. Add any new information you discover to the table you created in the activity.

Test

1. Name two reasons for data being collected online.
2. What are the two main reasons that websites use cookies?
3. Explain how cookies can be used to personalise a website. Give examples.
4. Why do people worry about the use of cookies on websites?

2.2 Danger online

In this lesson

You will learn:

- about the risks you face online from cybercriminals.

Spiral back

In Student Book 6 you learned how to behave responsibly online and to respect other internet users. Not every user on the internet behaves responsibly. Some users behave disrespectfully. Other users are involved in criminal activity. In this lesson and in Lesson 2.3 you will learn how to recognise criminal activity on the internet so that you can protect yourself.

Cybercrime

One risk you face on the internet is **cybercrime**. A **cybercriminal** is someone who uses the internet to commit their crimes. Some cybercriminals sell illegally over the internet. Some cybercriminals attempt to steal money and other possessions from people.

Cybercrime methods

Criminals use several methods to steal people's money and identities:

Identity theft – A criminal steals personal information such as a person's name and address. They also steal official information such as social security numbers or passport numbers. Criminals can use a stolen identity to pretend they are someone else. They can open a bank account or a credit card account in someone else's name. They then use these fake accounts to buy goods or to steal money from banks.

Phishing – A criminal creates a fake website that looks like the official website of a real bank or internet shopping website. The criminal then writes a fake email that looks like it comes from the bank or shopping website.

The email tells the person receiving the email that there is a problem with their account and they must log in to solve the problem. There is a link to the fake website. When the person logs in, the fake website records their username and password. The criminal can then use the person's login details to steal money.



Only a small percentage of people are fooled by phishing emails, but every victim can lose hundreds or thousands of dollars to the criminal.

Scamming – A criminal sends out an email that asks for money. Sometimes the email contains a threat that the victim owes money and will have to go to court if they do not pay the debt. Sometimes the email offers something for sale at a very low price or asks for a donation to a charity. Scammers also use social media sites.

If the criminal emails enough victims, someone will be fooled and will send money to the criminal.

Hacking – A criminal breaks into a computer system, usually to steal files or personal information. Hackers usually try to break into computer systems that hold a lot of personal information. Bank systems are an example of the type of system that hackers attack.



Did you know?

Banks and other organisations sometimes employ hackers to attempt to break into their systems. This can reveal weak spots where the system is insecure. The organisation can then fix the weak spots before a malicious hacker finds them.

Be creative

Create a poster that warns about the dangers of cybercrime. Use some or all of the key words from this lesson. Search the internet for a good image to illustrate your poster. Your poster can warn about all the dangers of cybercrime or it can focus on one type of cybercrime, such as identity theft.

Malware

Malware is software that a criminal installs on a computer. Some types of malware are designed to steal personal information. Other malware is designed to damage files or stop a computer from working properly. Some types of malware are:



- **Virus** – spreads and infects files that it meets. That makes a virus difficult to cure. Some viruses are relatively harmless – they make messages pop up on your screen, for example. Other viruses are more dangerous – they can destroy files and steal personal data.
- **Trojan** – malware that is hidden inside another program, for example in a game. When someone uses the program, the malware is released and starts to work. Trojans are popular with cybercriminals because they are easy to write and spread.
- **Ransomware** – encrypts all the data on a computer. That means that the data cannot be used. The victim of a ransomware attack receives a message demanding a payment to remove the encryption.
- **Spyware** – sits on a computer and records what the user is typing. Criminals can use spyware to discover login details.
- **Adware** – places unwanted adverts on a computer. The adverts are often displayed in another web page such as a social media or internet shopping site. Adware is not as damaging as other malware, but it can be annoying. If you have adware on your computer, it is a warning that your computer is vulnerable to other more dangerous malware.



Criminals often use email to spread malware. They often send fake emails with subjects that encourage you to open them. For example, the email subject might say you have won a prize in a competition or warn you that your computer has a serious problem you need to deal with.

The email encourages you to open or install a software application or open a document. The file contains malware, so when you open the file your computer is infected with the malware.



Did you know?

One of the most serious malware attacks was a ransomware attack called WannaCry. It affected over 200,000 computers in 150 countries. WannaCry closed important computer systems including some hospital systems, so the hospitals had to cancel operations. The cost of the attack was estimated at $4 to $8 million.

Activity

Read the Did you know? box. Search the internet to find out about other major malware attacks that have taken place. Choose one of the malware attacks that interests you. Write a brief description of the attack. Some things you can include in your description are:

- the name of the malware
- the type of malware
- the effect of the malware
- when the attack took place
- how many people were affected
- the cost of the attack
- the person or people responsible for the attack.

Extra challenge

Trojan malware is sometimes called 'Trojan horse' malware. Its name comes from an Ancient Greek story. Search the internet to find out what the original Trojan horse was. Why is the name Trojan horse used for this type of malware?



Test

1. Name three types of malware.
2. What is ransomware and why is it dangerous?
3. Explain why a stolen identity can be valuable to a criminal.
4. Describe how criminals use emails to spread malware.

2.3 Protect your computer

In this lesson

You will learn:

- how to protect your computer against malware and hackers.



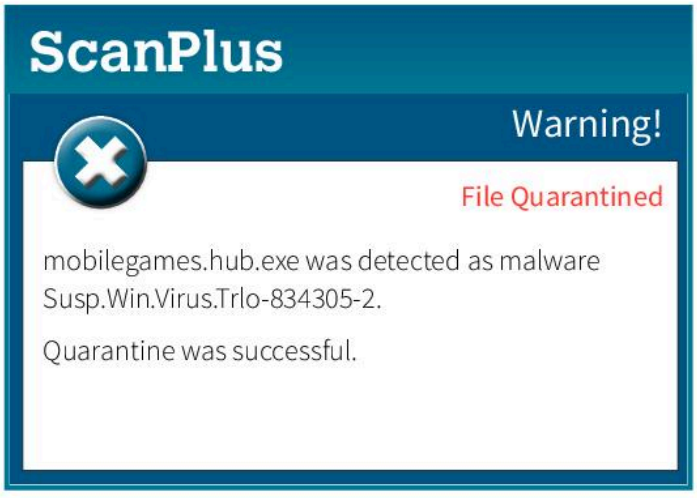
Anti-virus software

You can prevent malware from doing damage by installing **anti-virus software** (**AV software** for short) on your computer. AV software deals with all malware threats, not just viruses. AV software sits on your computer, looking out for malware.

How AV software works

Every piece of malware has a fingerprint. The fingerprint is a piece of programming code that is unique to the malware. This fingerprint is called a **signature**. AV software contains a database of malware signatures.

AV software continually checks the files on your computer, looking for malware signatures. If your AV software finds a malware signature in a file, it puts the file in **quarantine** – this means that the computer cannot open the file.



Modern AV software protects your computer very well. But there may be times when AV software fails to detect malware. If a new piece of malware is developed, your AV software will not recognise it because the signature of the malware will not be in the signature database.

AV software updates the signature database regularly. But there will always be times when your computer is vulnerable to new malware.

Activity

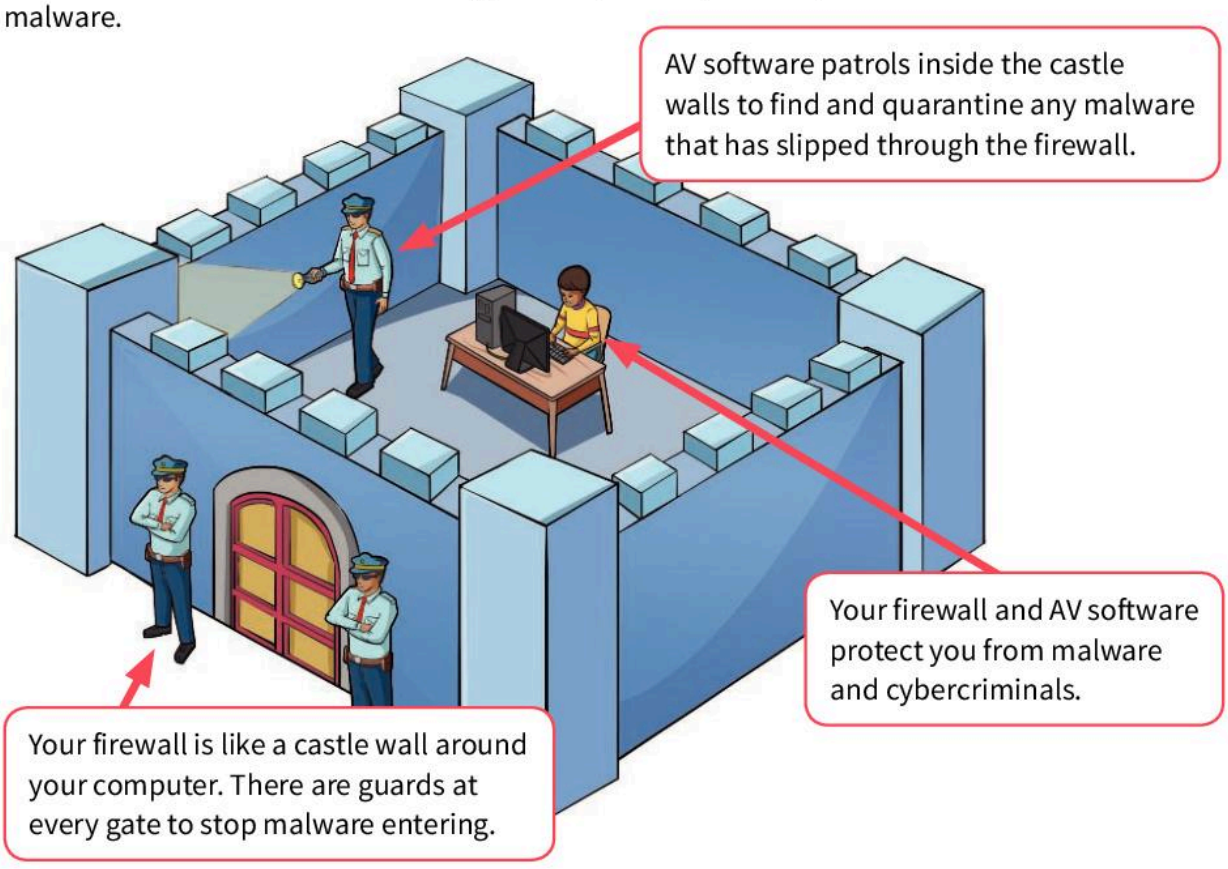
There are many companies that make AV software. Norton and Kaspersky are two examples. Search the web for the names of other AV software packages. Find reviews of different brands of AV software. Make a note of the good and bad points in the reviews for each product.

Firewall

AV software detects and quarantines files that contain malware. Another type of software that protects your computer is a **firewall**.

A firewall surrounds and protects your computer like the walls of a castle. A firewall checks all data before it allows the data to pass through the gates in the wall. Only data that comes from a safe source can pass through the wall.

Your firewall and AV software work together to protect your computer from malware.



Using your computer safely

You can improve your security by behaving responsibly and safely when you use your computer.

Use firewall and AV software

Never turn off the firewall or AV software on your computer. Never change any of the settings in the software. If you do, your computer is open to malware.

Use email safely

Malware can come from files attached to emails. If you open an email attachment, you may be inviting a criminal onto your computer.

Criminals and hackers use misleading email titles to tempt you to open a mail. For example:

"Congratulations! You have won first prize in our free draw."

"Warning! Your account is about to be closed. You MUST act now!"

Be wary of emails from people you do not know. Even emails that look genuine can be clever fakes. An honest email will never ask you to give away personal information or login details.

Spiral back



In Student Book 5 you learned about using strong passwords. Always use strong passwords that are difficult for a hacker to guess.

Use the web safely

When you are searching the web, only use sites that you trust. Files that you download can contain malware. Downloading apps and games is particularly risky. If you are downloading software, always use an official site.

When browsing the web, you will sometimes see messages that tell you there is a problem with your computer. The message will tell you that you need to download software to fix the problem. Do not download the software. Close the web page.

We have detected a problem with the virus protection on your computer.
You should scan for viruses immediately.
Download Virus Scanner

Never click on links in messages that appear when you are online.

Update your software

Keep the software on your computer up to date. Software companies constantly check their software for errors. Sometimes they find security errors. Criminals and hackers use security errors to access computers.

If your software provider sends you an update message, update your software immediately. The updated version will make it more difficult for criminals to install malware. If you think an update message may be malicious, check with an adult before accepting it.

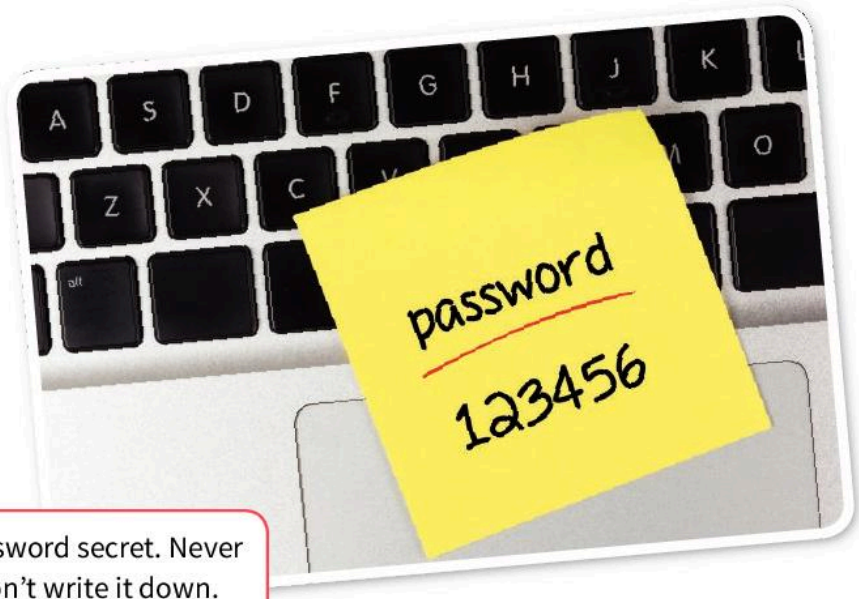


Your software provider will send update messages to a window on your screen, never as an email message.

Keep your password safe

Use a strong password that contains upper and lower case letters, numbers and other characters (for example, $). A strong password should be at least eight characters long

Change your password regularly. That makes it more difficult for hackers to access your computer even if they do manage to steal your password. If you think your password has been stolen, change it immediately.



Keep your password secret. Never share it and don't write it down.

Activity

Write a guide for students on how to work safely online and avoid the risk of malware.

Extra challenge

In Student Book 5 you learned the passphrase method to create strong passwords. Search the web to find another method for creating a strong password. Write down instructions with an example. Add these to your guide on working safely online.

Did you know?

Hackers use special software to crack passwords. A weak password that is 7 characters long can be cracked in less than 1 second. A strong password that is 12 characters long could take 200 years to crack.

Test

1. Name two types of software you can use to protect your computer from malware.
2. Why is it important to update the software on your computer regularly?
3. Explain how AV software and a firewall work together.
4. Explain why you need to be careful when you open a file attached to an email.

2.4 Cyberbullying

In this lesson

You will learn:

- that bullying can be a danger on the internet
- how cyberbullies work
- what to do if you experience cyberbullying.

What is cyberbullying?

A person who uses the internet to bully people is called a **cyberbully**. Cyberbullying is a serious problem. In some countries, over 50% of young people have experienced cyberbullying. Cyberbullying often takes place on social media sites, but bullies also use email and text messages.

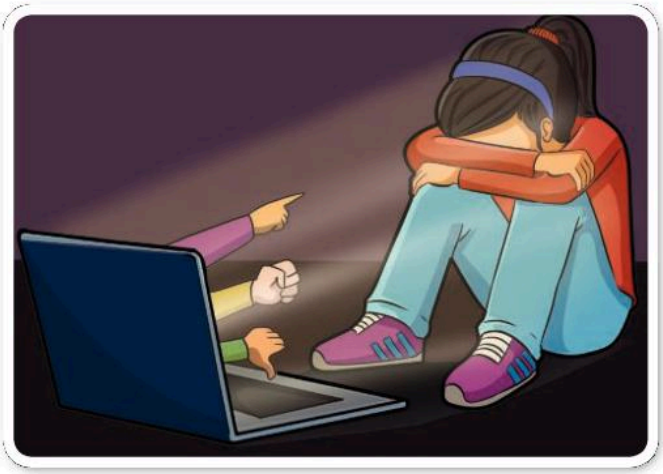
Methods cyberbullies use

- **Harassment** – A bully sends threatening messages by text message or instant messaging. This is a serious form of bullying. The messages often make physical threats. They can be very frightening.
- **Impersonation** – A bully chooses a target (a person they want to bully) and then sends messages that pretend to be from the target. The messages are sent to people who know the target. The aim is to cause trouble between the target and their friends.
- **Exclusion** – A bully sends messages that publicly exclude the target from social groups and events. For example, the bully posts a message on a social media site saying that the target is not invited to an event.
- **Dissing** – A bully circulates rumours (untrue stories and opinions) about a target. The rumours are designed to humiliate and ridicule the target.
- **Photographs** – Some cyberbullies use their smartphone camera to take photos of a target. They then share the photos by text message or social media. They sometimes use the photos in harassment messages.



Be creative

Write a short story about cyberbullying. Choose one of the methods that cyberbullies use. Describe an incident when a bully used that method. How did the target feel about the bullying? What did they do about it? Why did the bully act that way?



The effects of cyberbullying

Cyberbullying is malicious and persistent. It can seem impossible to escape from. A bully can strike at any time, using text messages and social media posts. The internet brings the bullying into the target's home.

Bullying over the internet is often anonymous. Not knowing who is carrying out the bullying makes it more frightening. Cyberbullying can be devastating if you are the target. It can affect every aspect of life.

Confidence

No matter how much a target tries to ignore cyberbullying, the bullying will affect their self-confidence. Low self-confidence can lead to anxiety and depression. Being the constant target of rumour and ridicule makes a person feel powerless and vulnerable.

Loneliness

Being bullied is a lonely experience. A person who is bullied often feels that no-one can help them. They feel isolated, which may cause difficulties in their relationships with family and friends. The target of cyberbullying can withdraw from their family and social groups.



Performance at school

A target of bullying sometimes does less well at school. The cyberbullying distracts them from their schoolwork and they find it difficult to concentrate in class. In extreme cases the person may start to miss classes. They no longer enjoy learning or being with school friends.

Change of personality

Being bullied can change someone's behaviour and personality. A person who is usually pleasant and respectful may start to misbehave or become aggressive. The stress caused by bullying can also lead to problems with sleeping and eating routines. The person may even become ill.

What makes a cyberbully?

If you are the target of a cyberbully, it is easy to think that you are to blame in some way for the bully's actions. It is important to realise that you are not to blame. There are many reasons why someone might become a cyberbully.

The cyberbully wants revenge for harm that has been done to them. It is rarely the person they bully who has caused the harm.

They turn their own unhappiness and insecurity on to someone else.

They want other people to like and admire them.



They are jealous of the person they bully.

Online bullying allows the bully to be anonymous so they believe they can get away with their bullying.

The cyberbully feels threatened by people who are different to themselves.

What should you do if you experience cyberbullying?

If you see someone being bullied

- Be careful not to join in. When you see a post that makes fun of someone, never click the 'like' button – not even as a joke. You will encourage the cyberbully and upset the person who the post is about.
- Say something positive about the person being bullied. Showing your support will make the bullying easier to bear.
- Offer support and friendship. Let the person talk if they need to. Encourage them to talk to an adult about the bullying.



If you are bullied

- Collect evidence. Take screenshots of posts so that you have a record even if the posts are deleted.
- Talk to someone you trust about the bullying. Talk to a friend, family member or teacher.
- Don't respond to the bullying. Responding can encourage the bully and make the situation worse.
- Spend some time offline.



Activity

Write an information sheet on how to survive cyberbullying. If you have the time to do the Extra challenge, use some of the information you discover in your web search.

Extra challenge

Search the internet to find one or more sites that give good information on how to deal with cyberbullying. Add the sites to your bookmark list.

Test

1. List the actions you should take if you are bullied online.
2. Explain two methods that a cyberbully uses.
3. Explain why someone might decide to bully another person online.
4. What should you do if you receive a threatening message on your phone or computer?



2.5 Use content responsibly

In this lesson

You will learn:

- what intellectual property is
- how to use internet content responsibly and legally.

In this unit you have learned about the dangers posed by criminals on the internet. Theft of money and identity theft are serious problems on the internet. Another problem is the theft of intellectual property.

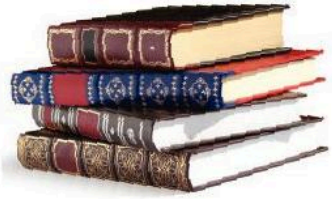


Intellectual property

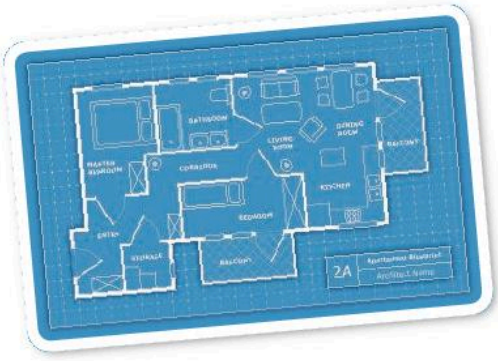
Intellectual property means something that you have created using your mind (your intellect) and that you own.

Intellectual property can be:

- written work – books, poems, articles and web pages
- images and artwork – cartoons, paintings, photos and sculptures
- music and songs
- plans and designs
- computer software and games.









Intellectual property rights protect the creator of a piece of work. It is illegal for another person to steal or misuse the work. You can only use another person's work if they give you permission.



Copyright, trademarks and patents

There are different types of intellectual property. Each has its own symbol.

Type of intellectual property	Symbol	Description
Copyright	©	**Copyright** means you have the right to copy your work. Other people must ask your permission before they can use it. You automatically have copyright for any work you create.
Trade mark	TM	Companies use trademarks to protect logos, slogans and product names. The Microsoft logo is a trademark.
Patent	℗	Inventors use patents to protect their new inventions. A patent stops other people from copying an invention and claiming it as their own idea.
Registered design	®	This is used to protect designs like wallpaper and carpet patterns.

Explore more

Every week, you use equipment and books at home and school. You use websites. You buy food and other items from shops. This week, look carefully at all the items you use and buy. Can you find an example of each of the four types of intellectual property described in the table?

What copyright means to you

You can legally use someone else's work in these situations:

- You have purchased their work. It may be an application program, game or music track. The work comes with a licence that allows you to use it. The licence will tell you exactly what you can and cannot do with the work.
- The owner of the work has given you permission to use their work. The owner will say what you can and cannot do with the content. For example, you may have permission to use a photo in your own work but not to change it.

Software piracy

Copyright theft is a type of cybercrime. Criminals make copies of music, games and films. They sell the copies, often over the internet. This type of crime is called piracy. When software is illegally copied and sold, it is called software piracy. When legal copies of software, music and games are sold, the person who created the work gets some money from each sale. When pirated copies are sold, the person does not get any money.

It is illegal to download and use pirated versions of software and other files.

Activity

Work with a partner or in a small group. Make a list of all the reasons why it might be a bad idea to download software from unofficial web pages.

Finding images on the internet

Many software apps provide images you can use – for example, web-editing tools like Wix and presentation software like Microsoft PowerPoint. You can also use many of the images that you find on the web.

Creative Commons

Many content owners make their images available free of charge on the internet. They use a special licence called **Creative Commons**. Creative Commons lets you use images without having to ask the owner for permission. Creative Commons music and video content is also available.

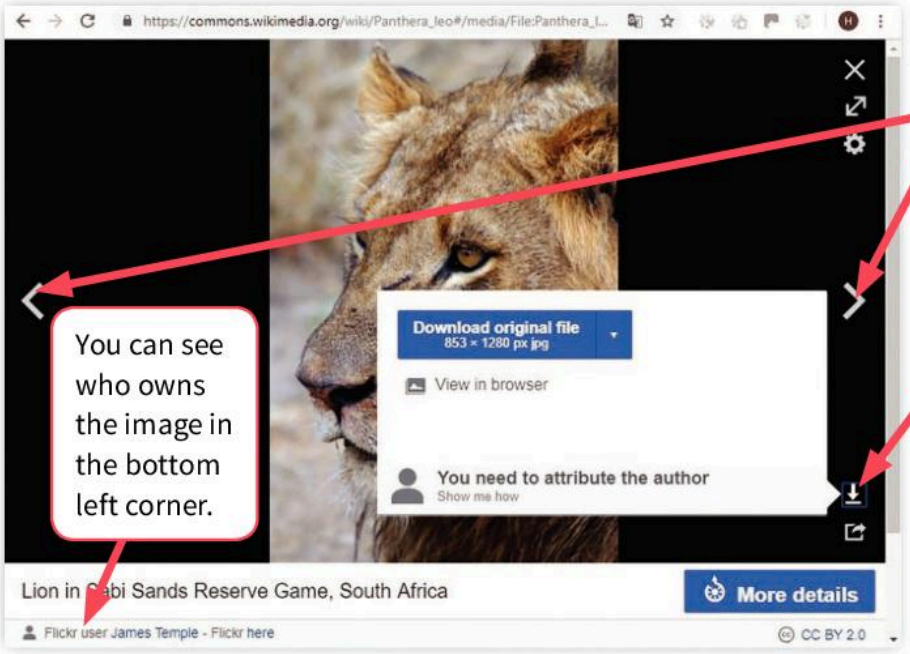
How to find Creative Commons images

There are websites that allow you to search for Creative Commons images. The one used in this example is called Wikimedia Commons. Other sites include Pixabay and Unsplash.

Enter key words for the images you are looking for in the search box. In this example the search is for 'Lions'. Your search will provide you with a list of images that match your key words. Sometimes you will see a page with information about the subject you have searched for. Scroll down the page to find the selection of images. Sometimes you will just see a list of links.

Clicking on an image opens a picture browser. The browser shows the image you have selected. There is information about the image below it.





You can see who owns the image in the bottom left corner.

The arrows at the left and right of the page let you move between images without returning to the search list.

Click on the download button. The download box reminds you that you must attribute (credit) the author.

Giving credit for the images you use

The owner of a Creative Commons image gives you permission to use the image. Most owners ask you to give credit for the work. A credit includes the owner's name and a link to the website you got the image from.

Below the message 'You need to attribute the author' is a link that says 'Show me how'. If you click the link, the website provides a line of text that you can use as a credit when you use the image.



Choosing a graphics download site

Make sure you use a site you can trust when you search for Creative Commons files. Data files you download from the internet can contain malware. If you are unsure about a site, ask an adult for advice before you download any files.

Activity

Search for an image of a tiger using Wikimedia Commons or another site that your teacher recommends. Download an image and add it to a word-processor document. Insert the credit provided by the website below the image.

Extra challenge

Search the web to find out what the legal penalties are for software piracy. Penalties vary in different countries. You may need to add the name of your country in your search string.

Test

1. Name two types of intellectual property.
2. What information should you use in a credit when you use someone else's work?
3. Explain why intellectual property theft and software piracy are bad for the owners of content.
4. Explain why it is a good idea to use a Creative Commons site when you are searching for images to use in your work.

2.6 Giving credit

In this lesson

You will learn:

- how to give credit when you use another person's work
- how to write a citation.

Using citations

How to credit an image

In the last lesson you learned how to find Creative Commons images on the web. Creative Commons images are shared by the people who made them. You can use most Creative Commons material free of charge, but you must give credit to the owner of the content. The text that you use to give credit is called a **citation** or an **attribution**.

In the lesson you downloaded an image from the Wikimedia Commons site. You learned how the site provides a citation for you to use with your downloaded image.

A Creative Commons citation includes the owner's name and a link to the image on the Wikimedia website. It looks like this:

> By Winfried Bruenken (Amrum) – Own work, CC BY-SA 2.5, https://commons.wikimedia.org/w/index.php?curid=1585973

Writing your own citation

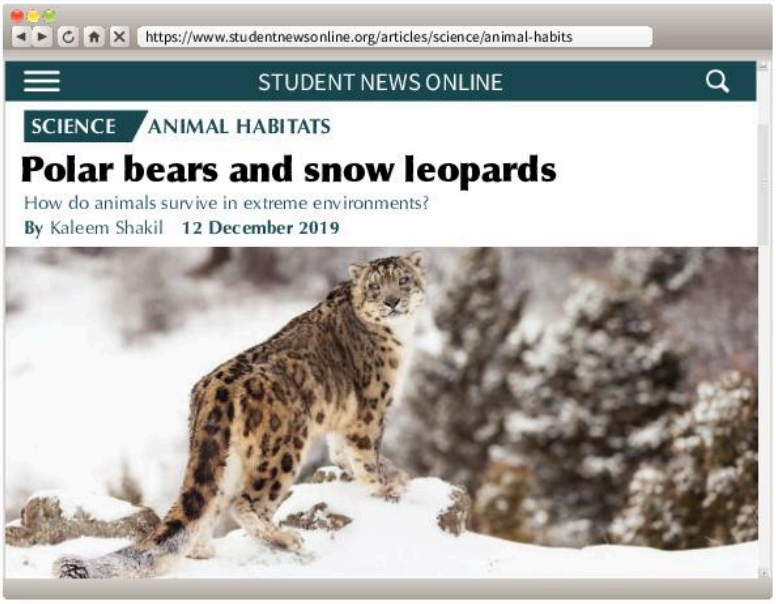
Sometimes the owner of a piece of work does not provide a citation for you to use. In that case, you need to write your own citation. There are four pieces of information you should include in a citation:

- **author:** the name of the person who created the work
- **year:** when the work was produced
- **title:** for example, the title of an article you are quoting from or the title of a picture you want to use
- **internet address:** the address of the site you are taking the work from. If the work you are using does not come from the internet, use the title of the book or newspaper where you found the work.



Can you find the four pieces of information you need for a citation in the news article at the top of the next page? Write a citation for this article.

There are times when you won't find some of the information you need. Don't worry. Just use what you can find.

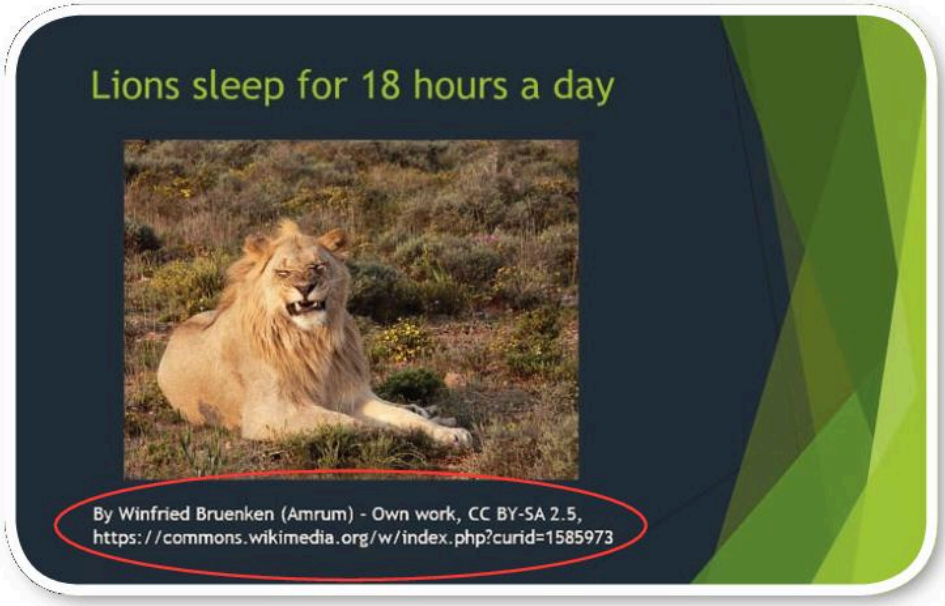


How to use a citation in your work

You should add a citation for every piece of content you use that is not your own work. Adding a citation is straightforward:

- Type or paste the citation immediately below the image or quoted text.
- Use a font that is a little smaller than the main text in your document. For example, if the text in your document is 12 points, use 10 points for your citation.

Here is an example of a citation in a presentation slide:



You can use the same method of adding a citation with other types of content. For example, you might use a video or a quote from a newspaper article. You might find a graph or a table that you want to use in your work. In every case, add a citation below the content.



Answer to the activity on page 54

A citation for the article shown at the top of page 55 will look like this:

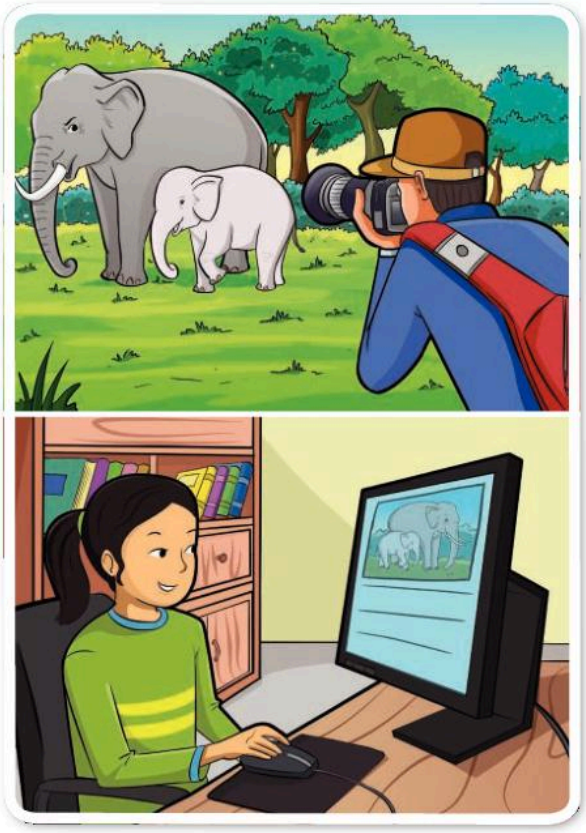
Kaleem Shakil, 2019, Polar bears and snow leopards, www.studentnewsonline.org

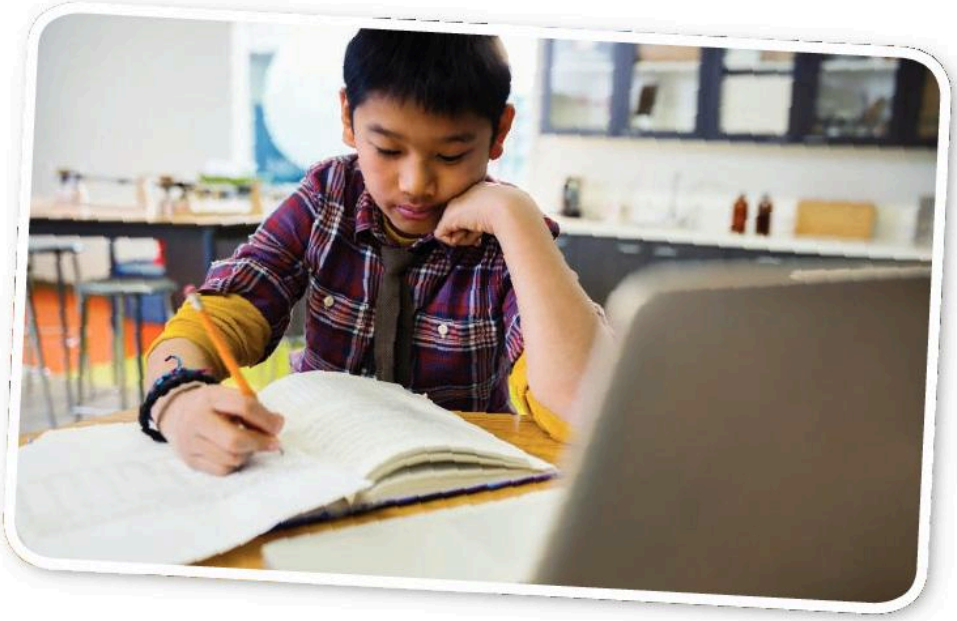
The citation includes the author's name, the year the article was written, the title of the article and the web address. Did you find all the information on the web page? If not, look at the image again to find the four parts of the citation.

Why are citations important?

There are four reasons why it is important to use citations to credit other people's work.

- You are giving credit to the person who created the content. Dishonest behaviour on the internet often involves someone losing money, for example by stealing someone's identity or their intellectual property. But people who create content are not just concerned with making money. People work hard to create new content and they are proud of it. They want to get credit for producing it.
- People use the internet to find information. They use web links from one document to another to find new facts. Someone who reads your work can follow the link in your citation to learn more about a subject.
- Citations make it clear what work is yours and what belongs to someone else. There is nothing wrong with using other people's knowledge in your own work. It is an important part of learning. But you must not pretend that someone else's work is your own. That is called **plagiarism**. If you use citations correctly, nobody can accuse you of plagiarism.
- Citations show your teacher that you have carried out research to complete your school assignments.





Digital citizen of the future



The internet can be dangerous. You can make it safer for yourself and others by doing the right things when you are online.

As a digital citizen of the future, remember to be cautious and to question everything you see and read on the internet. Keep your internet skills and knowledge up to date. Share your knowledge, and support friends and family who are learning how to use the internet.

Use the internet well and you will enjoy its benefits as you learn, work and lead your everyday life. Creating e-learning materials is a job that combines creative work with computing skills. Is it a career you would consider in the future?

Activity

Use word-processing or presentation software to create a short text about one of your hobbies or interests.

Write a paragraph about why you like this hobby or interest.

Find a relevant Creative Commons image to illustrate your text.

Search the web to find an interesting fact or quotation about your hobby or interest.

Add citations for the image and the quotation or interesting fact.

Extra challenge

When you download images and videos from the web, you must be confident that the site you are downloading from is safe and that the content will be free from malware. In the last lesson, two alternative sites to Wikimedia Commons were mentioned: Pixabay and Unsplash. Choose one of these and search the web for reviews of your chosen site.

Test

1 Why is it a good idea to search for images on a Creative Commons website?
2 What must you do if you want to use another person's content in an assignment?
3 What is plagiarism?
4 How can using citations improve your work?

Check what you know

You have learned

- how to recognise the risks and dangers on the internet
- how to avoid risks and dangers on the internet
- how to use internet content responsibly.

Try the test and activities. They will help you to see how much you understand.

Activities

Create a guide for students called 'Protecting yourself online'. You can present your guide as a word-processed document, as a presentation with slides or as a series of web pages.

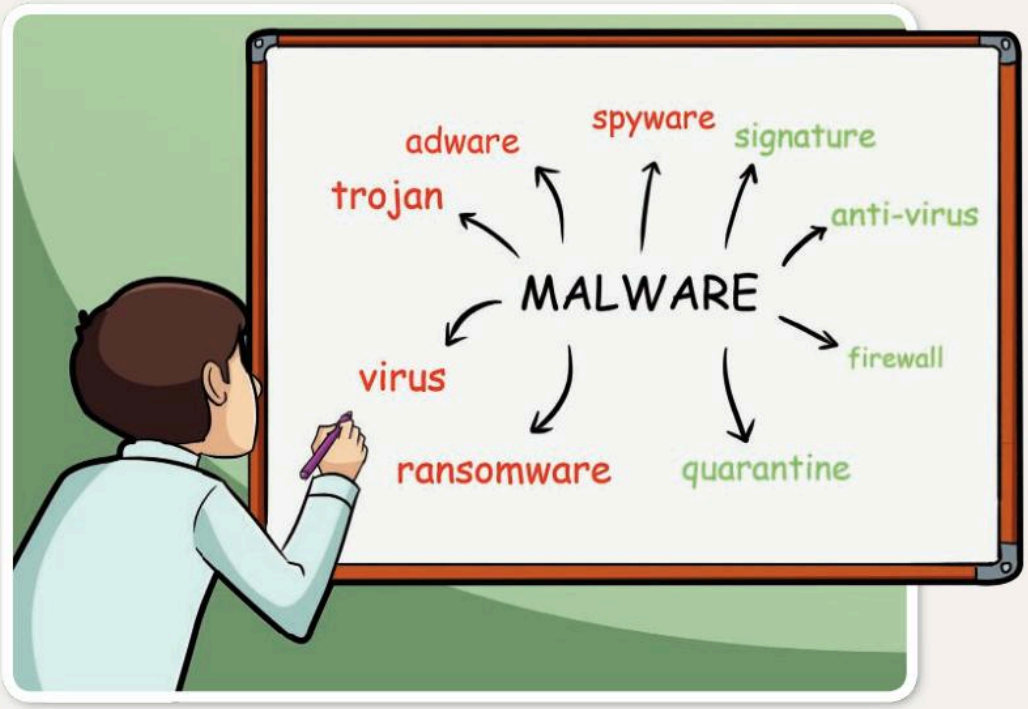
1 Say why it is important to protect yourself online.

2 Explain some risks people face online. You can include cyberbullying, malware or other risks you know about.

3 Add information about how people can protect themselves from risks online. For example, they can install anti-virus software and keep it up to date.

You can use the information in this unit and search online to find facts and images to illustrate your guide.

Remember to add citations for any quotations and images that you use.



Test

In this unit you have used websites. This test is about the websites you have used.

1 Think of one website that you have used to get information.
   - **a** Write down the name of the website.
   - **b** Write down one piece of information you got from this website.

2 Think of one online form that you have seen.
   - **a** Describe the online form.
   - **b** Write down one item of data collected by the online form.

3 What is a website cookie? What are cookies used for?

4 You can include information from a website in your work, but you must say where you got it from. Explain how to credit another person's work.

5 Explain how cookies can make websites easier to use.

6 Explain why intellectual property rights are important to a photographer.

Self-evaluation

- I answered test questions 1 and 2.
- I completed activity 1. I created a guide with reasons why it is important to protect yourself online.
- I answered test questions 1–4.
- I completed activities 1 and 2. I explained some of the risks people face online in my guide.
- I answered all the test questions.
- I completed all the activities. I described how people can protect themselves from risks online.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?